



3D Flexagon

Assembly Instructions

Lew Lefton, Georgia Tech School of Math, Decatur Makers

Version 1.0 February 2026



These instructions show how to make a 3D flexagon inspired by Martin Gardner's well-known hexaflexagons, which he first explored in his Mathematical Games column in the December 1956 issue of Scientific American. The design of this object is not mine. However, this approach may be new, since it is intended to be very accessible by using everyday household materials.

Materials needed

- 12 standard #1 metal paper clips
- 1 cocktail straw
- small pliers or other wire tool for cutting and bending wire *
- scissors or exacto knife to cut cocktail straw *

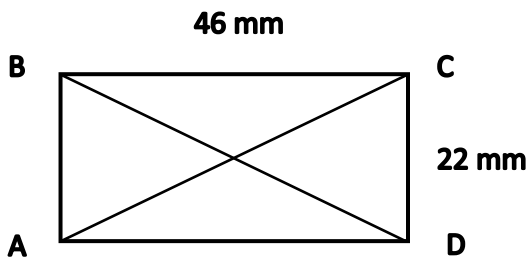
(*) Kits provided in the Gathering for Gardner 16 (G4G16) gift exchange have necessary wires and straw segments pre-measured and pre-cut, so you can skip steps 1 and 2.

Step 1. Unbend the paper clips to create 12 wires that are more or less straight. Cut 6 of these wires to length 95 mm and 6 of them to length 85 mm. You can use the figures below to measure lengths.

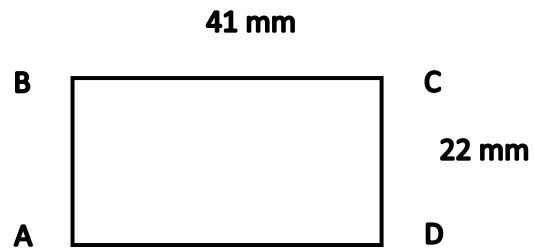
Step 2. Cut the straw into 6 small tubes of length 22 mm. You can use the figures below to measure lengths.

95 mm 
85 mm 
22 mm 

Step 3. The 12 wires are going to be bent in a very specific way. The six 95 mm wires will end up with angles of $\sim 60^\circ$, and the six 85 mm wires will end up with angles of $\sim 90^\circ$. Use the following figures as a guide when bending the wires.



~ 60° template for 95 mm wires



~ 90° template for 85 mm wires

Three of the six 95 mm wires will be bent into a 3D shape with ~60° angles, and the other three 95 mm wires will be bent into a mirror image of that shape. **We will label these two shapes 60L and 60R.**

Similarly, three of the six 85 mm wires will be bent into a 3D shape with ~90° angles, and the other three 85 mm wires will be bent into a mirror image of that shape. **We will label these two shapes 90L and 90R.**

Step 3.1 - Bending a 60L and 60R wires:

For a 60L wire, place one end of a 95 mm wire on the lower left corner (A) of the 60° template, so that the bottom 22 mm of the wire is lined up with the AB edge of the figure. Bend the wire, flat along the paper, **clockwise** until it lines up with the D vertex. This creates the first angle of ~ 60°. Now bend the remaining 22 mm segment of the wire up, **away from the paper**, toward the B vertex, until it makes another angle of ~ 60°. Repeat this two more times to create three “60L shapes”. See Figure 1.



Figure 1: Six 95 mm wires bent into mirror images 60L and 60R.

For a 60R wire, place one end of a 95 mm wire on the lower right corner (D) of the 60° template, so that the bottom 22 mm of the wire is lined up with the DC edge of the figure. Bend the wire, flat along the paper, **counterclockwise** until it lines up with the A vertex. This creates the first angle of ~ 60°. Now bend the remaining 22 mm segment of the wire up, **away from the**

paper, toward the C vertex, until it makes another angle of $\sim 60^\circ$. Repeat this two more times to create three “60R shapes”. See Figure 1.

Step 3.2 - Bending a 90L and 90R wires:

For a 90L wire, place one end of a 95 mm wire on the lower left corner (A) of the 90° template, so that the bottom 22 mm of the wire is lined up with the AB edge of the figure. Bend the wire, flat along the paper, **clockwise** until it lines up with the C vertex. This creates the first angle of $\sim 90^\circ$. Now bend the remaining 22 mm segment of the wire up, **away from the paper**, toward the B vertex, until it makes another angle of $\sim 90^\circ$. This final segment should be approximately normal to the plane of the paper. Repeat this two more times to create three “90L shapes”. See Figure 2.



Figure 2: Six 85 mm wires bent into mirror images 90R and 90L.

For a 90R wire, place one end of a 95 mm wire on the lower right corner (D) of the 90° template, so that the bottom 22 mm of the wire is lined up with the DC edge of the figure. Bend the wire, flat along the paper, **counterclockwise** until it lines up with the B vertex. This creates the first angle of $\sim 90^\circ$. Now bend the remaining 22 mm segment of the wire up, **away from the paper**, toward the C vertex, until it makes another angle of $\sim 90^\circ$. This final segment should be approximately normal to the plane of the paper. Repeat this two more times to create three “90R shapes”. See Figure 2.

Step 4. Assembly.

For clarity, lay out each wire segment on a sheet of paper, and mark the paper to make it easier to identify them. Start with a 60L and 90L wire and two tubes to make a **basic shape**, as shown in Figure 3.

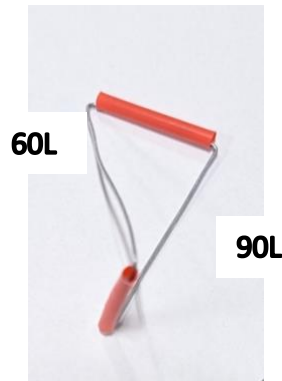


Figure 3: Basic shape using two tubes and a 60L, 90L segment.

Add a 60R segment and a 90R segment with a third tube to create a **basic double shape**, as shown in Figure 4.

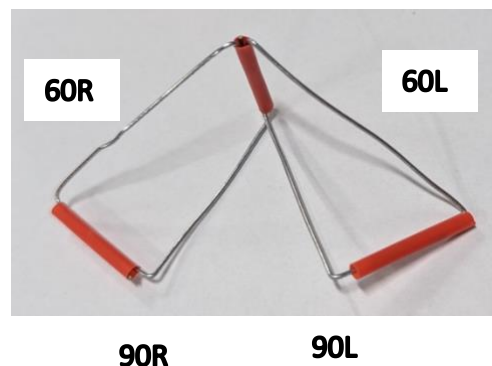


Figure 4: Basic double shape using 60L, 90L, 60R and 90R segments.

Note 1: There are *four wires going into the center vertical tube, two from each direction*. When fully assembled, *that will be true for all the tubes*.

Note 2: The 90L and 90R wires will end up making a large equilateral triangle (each side has two wire segments) that lies flat with “horizontal” tubes pointing outward. The 60L and 60R wires connect the “vertical” tubes to the “horizontal” tubes in an “up and down” pattern. See the finished product in Figure 8.

Now continue the pattern. Going clockwise, add a 90R and 60R wire to the 90L and 60L side of the basic shape, see Figure 5.

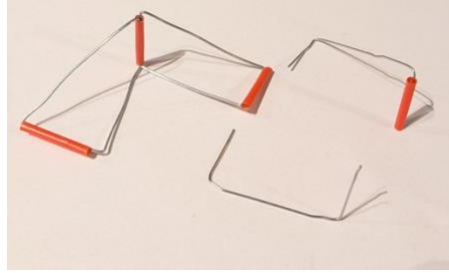


Figure 5: Adding 60R and 90R segments to the 90L and 60L side of the basic shape.

Continuing the pattern, going clockwise, add a 90L and 60L wire, see Figure 6.

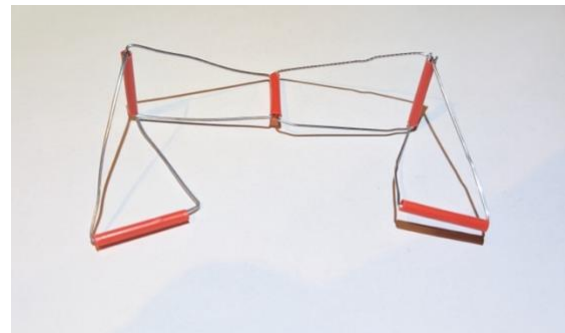
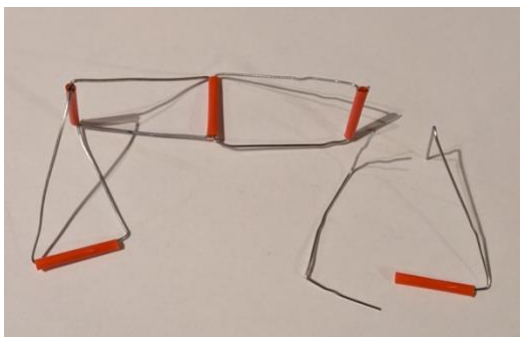


Figure 6: Continuing clockwise, adding 90L and 60L wires (left image). After attached (right image).

Almost done! Take the remaining tube and the remaining 60L and 60R wires and connect them to each other with the final tube as shown in Figure 7. Then complete the flexagon with the two remaining 90L and 90R wires, as shown in Figure 7. The finished 3D Flexagon is in Figure 8.

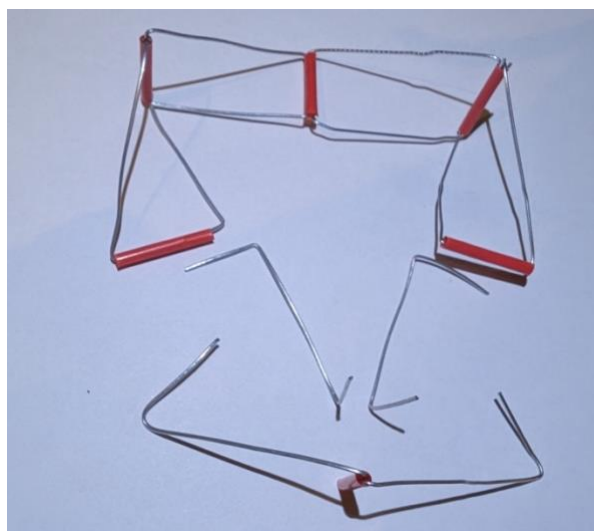


Figure 7: Connect the 60L and 60R wires with the final tube, and attach to complete the flexagon.

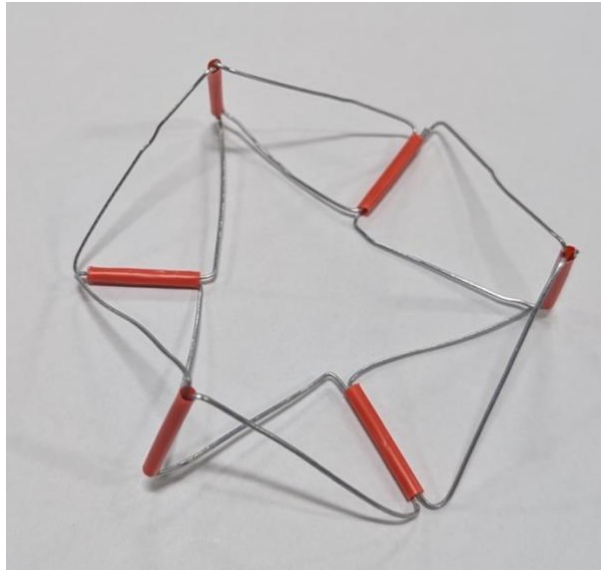


Figure 8: Finished 3D Flexagon

Good Job! Time to Flex.

Flex by repeatedly pulling horizontal segments into a vertical position. How many distinct configurations can you see?

Depending on how the wire was cut and bent, the Flexagon may need be adjusted by making small bends in the wire to loosen or tighten angles. You may also remove and trim the ends of the wire if necessary.